MallAI SOP and Infrastructure Design

Edgistify Innovation Challenge 2025

Team Name: SwiftChain

This document outlines the Standard Operating Procedures (SOPs) and Infrastructure Design for MallAI, focusing on efficient operations in both physical visit and online order fulfillment modes. It also details the manpower allocation and shift planning required to support these operations.

Standard Operating Procedures (SOPs)

The following SOPs detail the processes for handling customer interactions and order fulfillment in two distinct modes: Physical Visit Mode and Online Order Mode.

Physical Visit Mode

1. **Customer Entry:** Greet customer upon entry with a friendly welcome.
2. **Product Inquiry:** Assist customers with product information and availability.
3. **Product Selection:** Guide customers to the desired products, offering alternatives or recommendations.
4. **Checkout Process:**
   - Scan items and confirm the total amount.
   - Process payment via accepted methods (cash, card, mobile payment).
   - Bag the purchased items securely.
5. **Customer Exit:** Thank the customer for their purchase and invite them to return.

Online Order Mode

1. **Order Receipt:** Receive online order notification with order details.
2. **Order Verification:** Confirm product availability and order accuracy.
3. **Fulfillment Process:**
   - Retrieve items from the backroom fulfillment area.
   - Carefully pack items ensuring appropriate packaging.
4. **Shipping & Delivery:**
   - Prepare the package for shipping with a tracking number.

    - Coordinate with delivery service for pickup and delivery.
5. **Order Confirmation:** Send order confirmation and tracking information to the customer.

Infrastructure Layout

The store is zoned into two primary areas to facilitate efficient operation:

- **Front:** Customer-facing area for product display and sales.
- **Backroom Fulfillment:** Area for inventory storage, order processing, and packing.

Manpower and Shift Plan

The following tables outline the necessary roles and shift plans for MallAI operations. Clear role definitions and structured shift schedules ensure optimal staffing and service coverage.

Roles and Responsibilities

Role	Responsibility	Count
Sales Associate	Assisting customers, managing checkout, maintaining store appearance.	3
Fulfillment Specialist	Processing online orders, managing inventory, coordinating shipments.	2
Store Manager	Overseeing store operations, managing staff, ensuring customer satisfaction.	1

Shift Plan

Time	Activity	Staff Allocation
9:00 AM - 1:00 PM	Opening store, assisting early customers, processing online orders.	2 Sales Associates, 1 Fulfillment Specialist
1:00 PM - 5:00 PM	Peak customer traffic, managing inventory, handling online order surge.	3 Sales Associates, 2 Fulfillment Specialists, Store Manager
5:00 PM - 9:00 PM	Closing store, assisting late customers, finalizing online orders, store cleaning.	2 Sales Associates, 1 Fulfillment Specialist, Store Manager

Conclusion

This SOP and Infrastructure Design document provides a foundational framework for MallAI's operations. By adhering to these procedures and utilizing the proposed infrastructure layout, MallAI can effectively manage both physical and online sales channels, ensuring customer satisfaction and operational efficiency. The detailed manpower and shift plan will ensure adequate staffing coverage throughout the day.